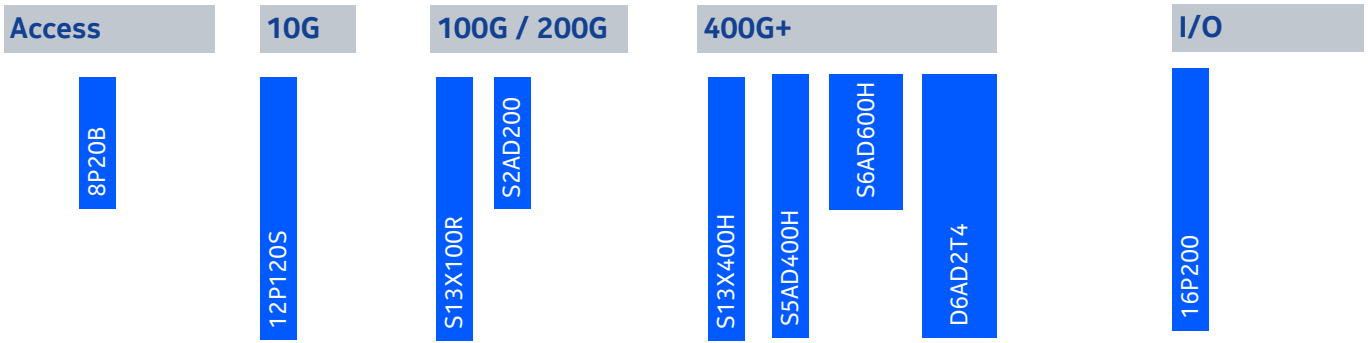
1830PSS Product Overview

Optical Transponders

- Optical Networks
- v43 – May 2024

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Transponder Summary



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Access / Metro

Unit Name	Part #	Ntwk Rate	Ntwk Ports	Client Ports	Client Srv	Size (H / W)	Supported Chassis	Avail
8P20B 10G Dual 8xAny Muxponder	3KC49272AA	10G	2 x SFP+	8 x SFP	1 GE OC3/STM1 OC12/STM4 OC48/STM16 FC-100/200/400 HD-SDI / SD-SDI OTU1	Half/ 1-slot	PSS-4 PSS-8 PSS-16ii PSS-32	GA

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10G Transponders / Muxponders

Unit Name	Part #	Ntwk Rate	Ntwk Ports	Client Ports	Client Srv	Size (H / W)	Supported Chassis	Avail
12P120s High Density 10G Transponder	3KC49081AA	10G	6 x SFP+	6 x SFP+	10GE OTU-2/2e OC192/STM64 FC-800/1200	Full / 1-slot	PSS-4 PSS-8 PSS-16ii PSS-32	GA

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+100G Transponder Naming

Line Ports
- S single WDM line port
- D dual WDM line ports

Client Ports
- # of client ports

Transponder Type
- X muxponder (internal OTN sw)
- AD add/drop

Payload Capacity (Gb/s)
- Example: 600Gb capacity

S 6 X 600 H



S6AD600H

Variant
- H high performance / LH
- R Regional
- L L-band
- E Encryption

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100G & Above Transponders

Unit Name	Part #	Ntwk Rate	Ntwk Ports	Client Ports	Client Srv	Size (H / W)	Supported Chassis	Avail
S13X100R **S13X100E** 100G Muxponder	8DG63207AA 8DG63988AA	100G	1	CFP4 QSFP28/ QSFP+ SFP+	10GE 40GE / OTU3 100GE / OTU4 OTU-2/2e OC-192/STM64 FC-800/1200/1600	Full / 1-slot	PSS-8 PSS-16II PSS-32	GA
S2AD200H 200G Muxponder	3KC69993AA	100G / 200G	1	2 x QSFP28	100GE, OTU4	Half / 1-slot	PSS-8 PSS-16II PSS-32	GA
S13X400H 400G Muxponder	3KC72184AA	400G CFP2-DCO	1	QSFP28 QSFP28/QSFP56 SFP/SFP+	10GE/25GE 100GE/OTU4 400GE OTU-2/2e FC-16/32	Full / 1-slot	PSS-8 PSS-16II	2H ‘24
S5AD400H 400G Muxponder	3KC70803AA	100G – 400G CFP2-DCO	1	4xQSFP28 1xQSFP56-DD	100GE / OTU4 400GE	Full / 1-slot	PSS-8 PSS-16II PSS-32	GA
S6AD600H / **S6AD600L /S6AD600E** 600G Muxponder	3KC71192AA 3KC71722AA 3KC71270AA	100G – 600G	1	5xQSFP28 1xQSFP56-DD	100GE / OTU4 400GE / OTUCn	Half / 2-slot	PSS-8 PSS-16II PSS-32	GA
D6AD2T4 / D6AD2T4L 2.4T Muxponder	3KC71964AA 3KC72384AA	up to 1.2T	2	4xQSFP56-DD 2xQSFP56-DD/ QSFP-DD800	400GE 800GE	Full / 2-slot	PSS-16II	1H ’24 / 2H ‘24

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1830 PSS Aggregation I/O Cards

Ports
Total # of ports

Port Type
- P programmable (10GE, SONET/SDH, Fibre chan)
- UD Uplink, distributed switching

Card Capacity (Gb/s)
Example: 200Gb capacity

16 P 200



16P200

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I/O Units

Unit Name	Part #	Port Rate	I/O Ports	Optics	Services	Size (H / W)	Supported Chassis	Avail
16P200 16 port Multi-Rate I/O	3KC71203AA	10G – 100G	12 x SFP+/SFP28 2 x QSFP28 client 2 x QSFP28 line	Grey	10GE, OTU-2/2e OC-192 / STM-64 8/10/16/32 FC 100GE	Full / 1-slot	PSS-8 PSS-16ii PSS-32	GA

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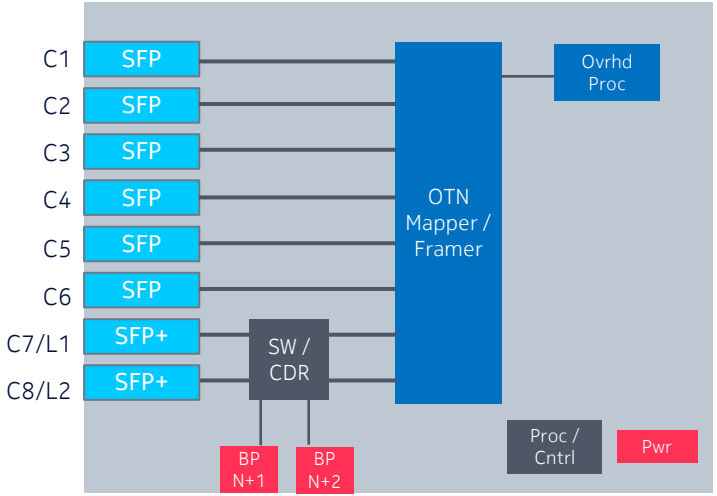
8P20B – 10G Dual 8xAny Muxponder

Unit Name	8P20B
Part #	3KC49272AA
Size	Half Height, 1-slot



8P20

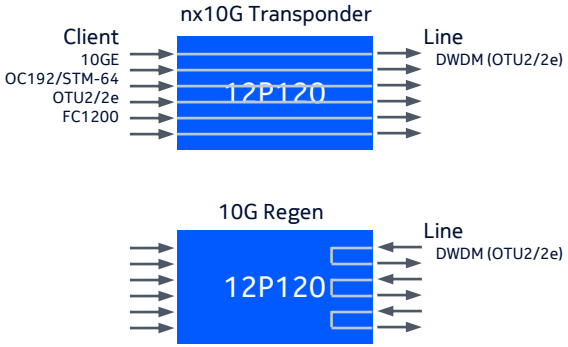
- Key Features:
  - Low cost 8xAny Muxponder with dual 10G line port
  - Muxponder or ADM on a blade applications
  - 2 x 10G SFP+ (OTU2/2e) network ports
  - 8 x SFP client ports
  - Client services: 1 GE, OC-3/12/48, STM-1/4/16
  - Protection: ODU0/1 SNCP
  - Line FEC: RSFEC / EFEC / None
  - Optional client/line mated with 11QPA4B
- OAM&P
  - GCC0/GCC1/GCC2 communications channel
  - Facility / Terminal Loopbacks
  - Optical, OTN, client Ethernet PMs



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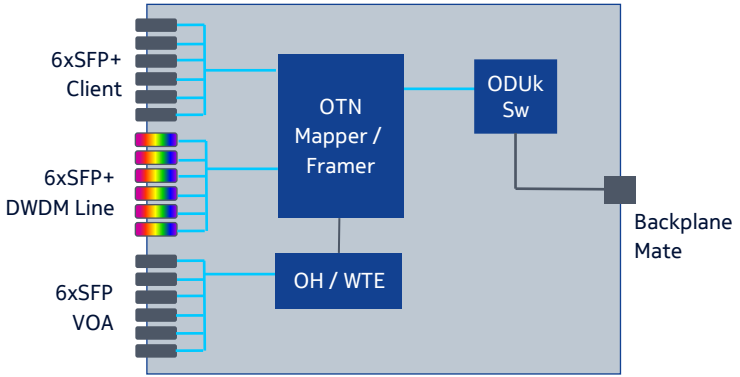
12P120S – High Density 10G Transponder

Unit Name	12P120S
Part #	3KC49081AA
Size	Full Height, 1-slot




12P120

- Key Features:
  - High Density n x10G Transponder
  - Integrated OTN switching enables sub-ODU0/1 grooming
  - Standalone Transponder or Interconnected Mate mode (paired)
  - 6 x 10G (SFP+) network ports
  - 6 x 10G (SFP+) client ports
  - Client services: 10 GE, OC-192/STM64, OTU-2/2e, FC-800/1200
  - Line FEC: RSFEC / EFEC / EFEC2 / None
- OAM&P
  - Facility / Terminal Loopbacks
  - Optical, OTN, client PMs



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S13X100R 100G Muxponder

Unit Name	S13X100R / S13X100E
Part #	8DG63207AA / 8DG63988AA

	Features	
Line	• 1 x 100G WDM line port • Nokia PSE-2c coherent DSP • SD-FEC (25%)	
Client	• 10 x SFP+	10GE OTU-2/2e OC-192/STM64 FC-800/1200/1600
	• 1 x QSFP+	40GE / OTU3
	• 1 x QSFP+/QSFP28	40GE / OTU3 100GE / OTU4
	• 1 x CFP4	100GE / OTU4
Features	• 100G Multi-service Muxponder supporting10G, 40G, 100G clients in single card • Integrated OTN switch (ODU0/1/2/ flex) • Optional L-Band version • Optional Encryption version	
Protection	• Y-cable / OPSB client protection • OCH protection (OPSA)	
Power	• ~ 162 W (typical)	

NOKIA
PSE-2c



S13X100

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S13X100 Using PSE-2C

Enabling dense, highly integrated 100G optics

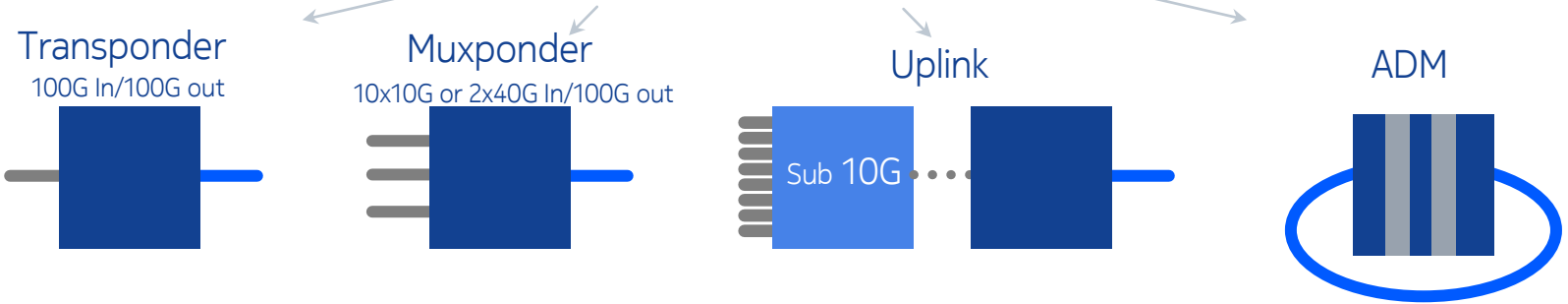
S13X100R



Low Power
High Integration

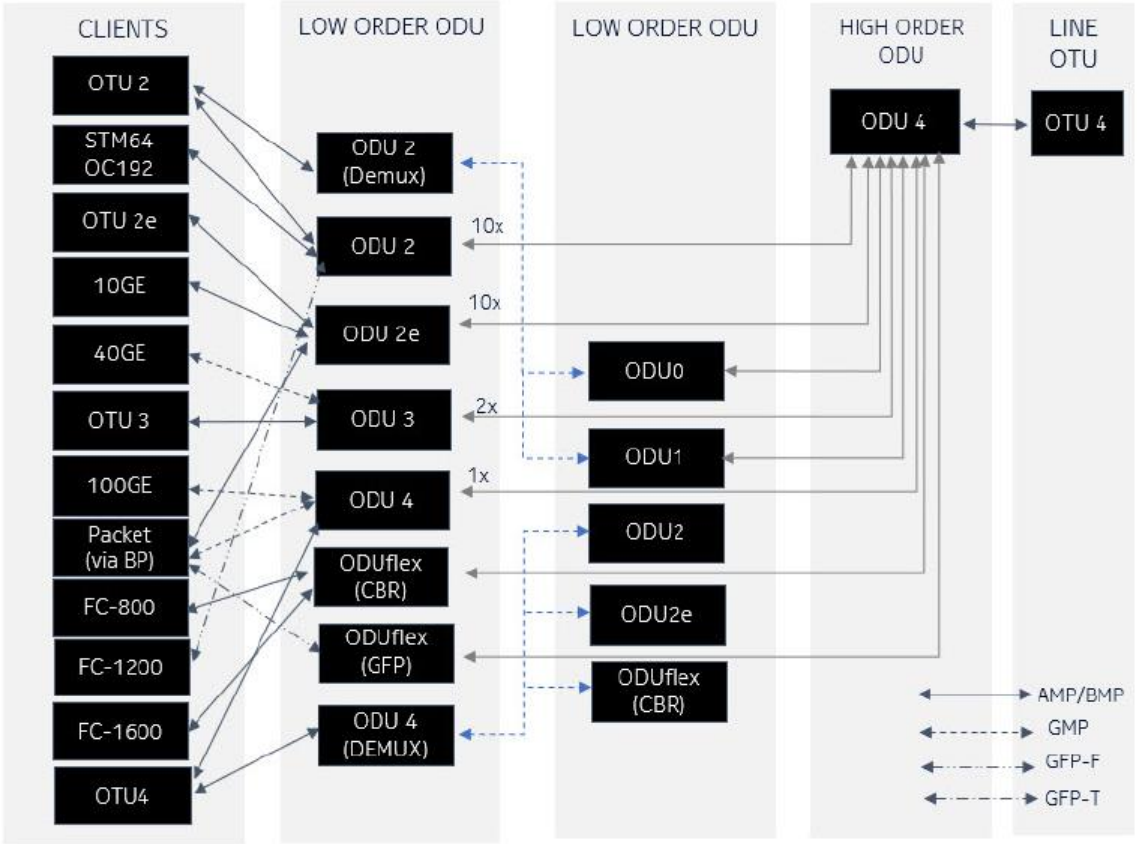


Flexible Clients
100G WDM Optics



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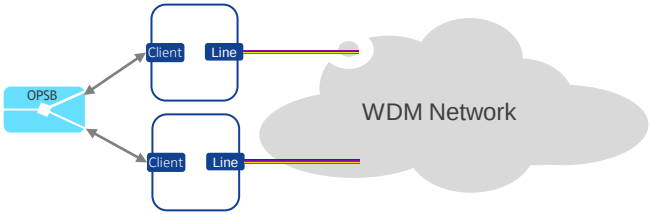
S13X100 OTN Service Mapping



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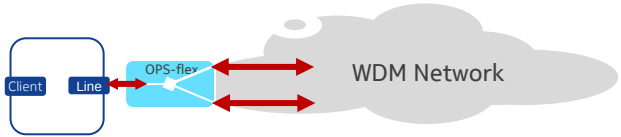
Protection Options on the S13X100

1. **OSNCP Protected Client, Protected Line card (OPSB)**



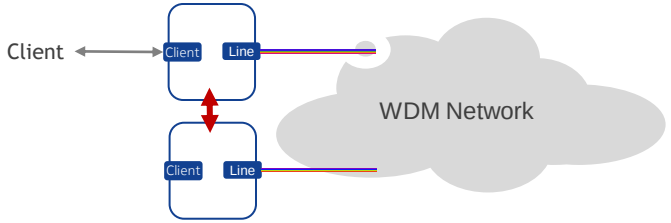
Multi-Shelf/Same-Shelf

2. **OCH Protected Line Card Conventional OADM (OPS-Flex)**



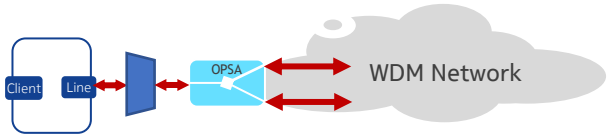
Conventional ROADM, TOADM, & FOADM

3. **ESNCP Protected Client, Protected Line card**



Same Shelf, Adjacent Slot

4. **Optical Multiplex Section Protected (OMS-P) Conventional OADM (OPSA)**



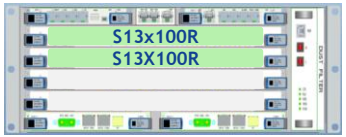
Conventional ROADM, TOADM, & FOADM
Requires matched filters on both ends

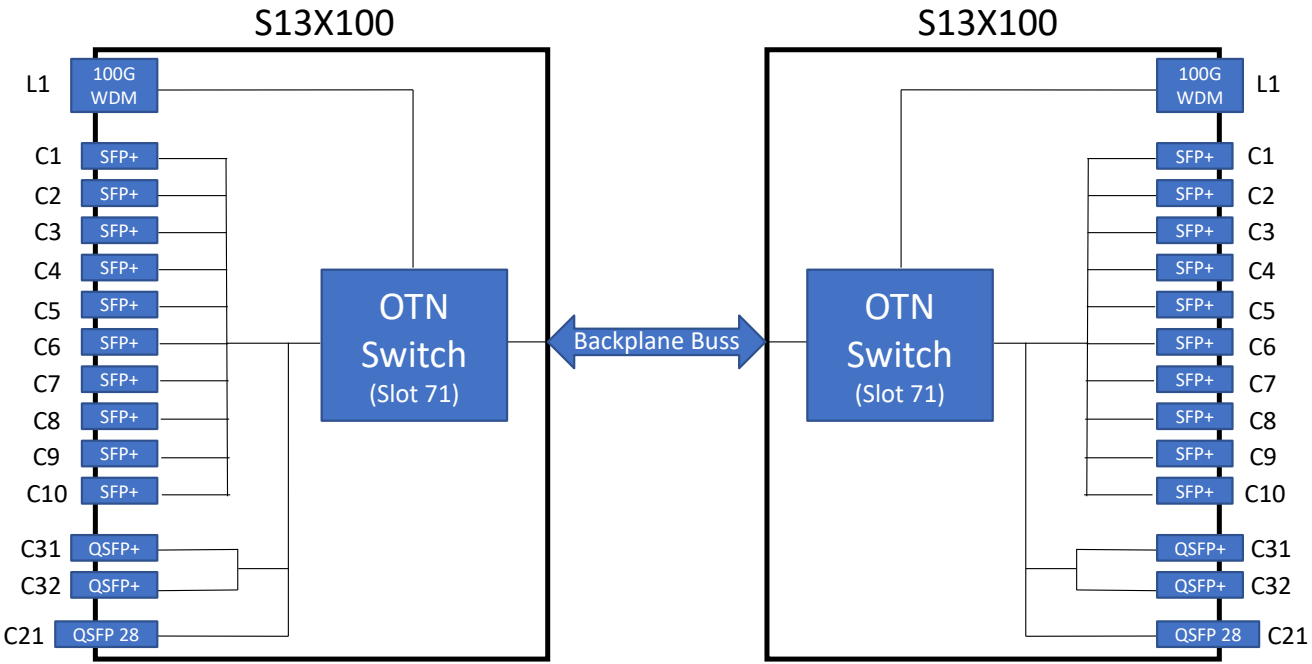
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The Secret Sauce: S13X100 embedded OTN Switch with Backplane Buss

Enabling ODU-0/1/2/3/4 Grooming and switching from Line-Line and Client to Line

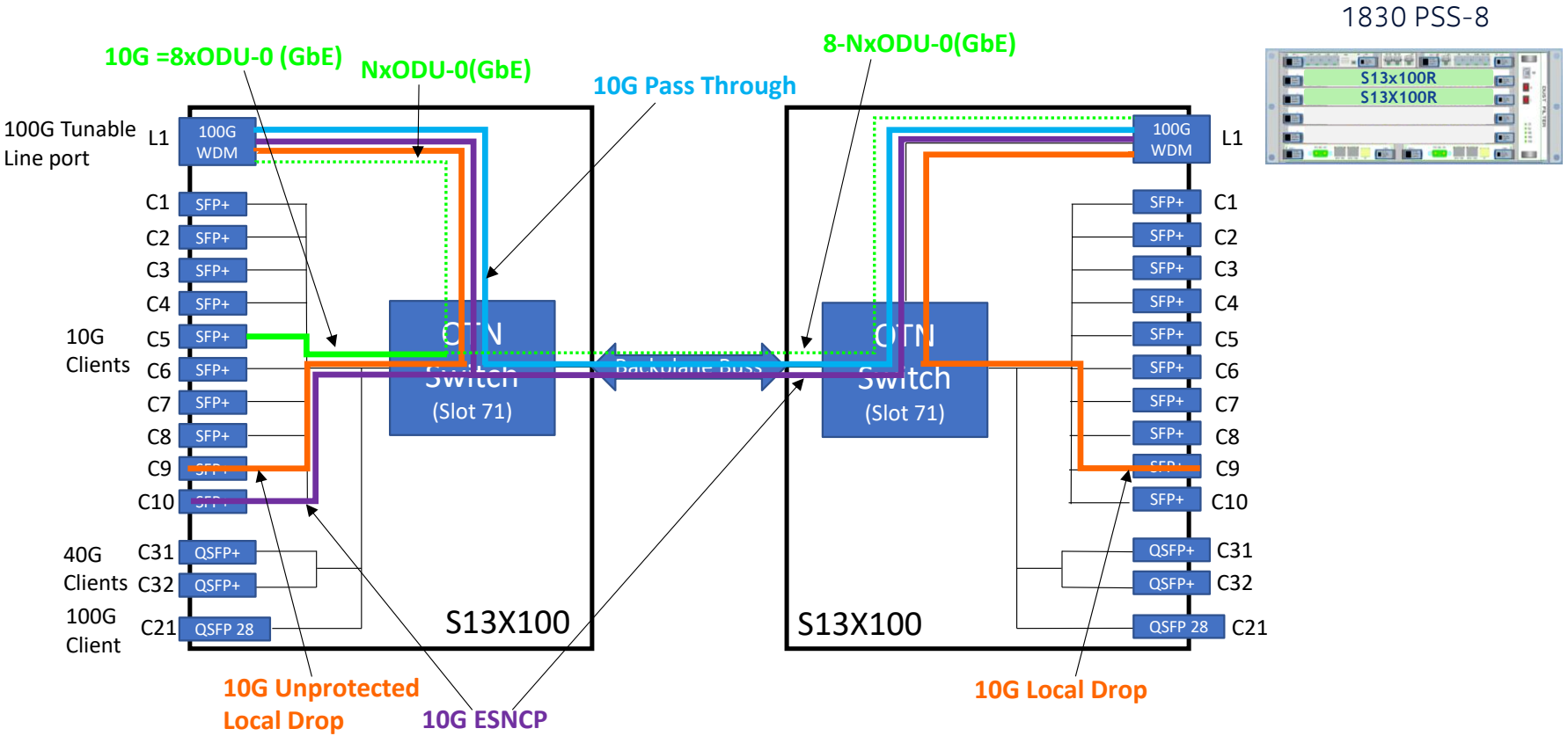
1830 PSS-8





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S13X100 10G and Sub-Rate Grooming examples

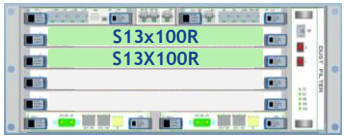


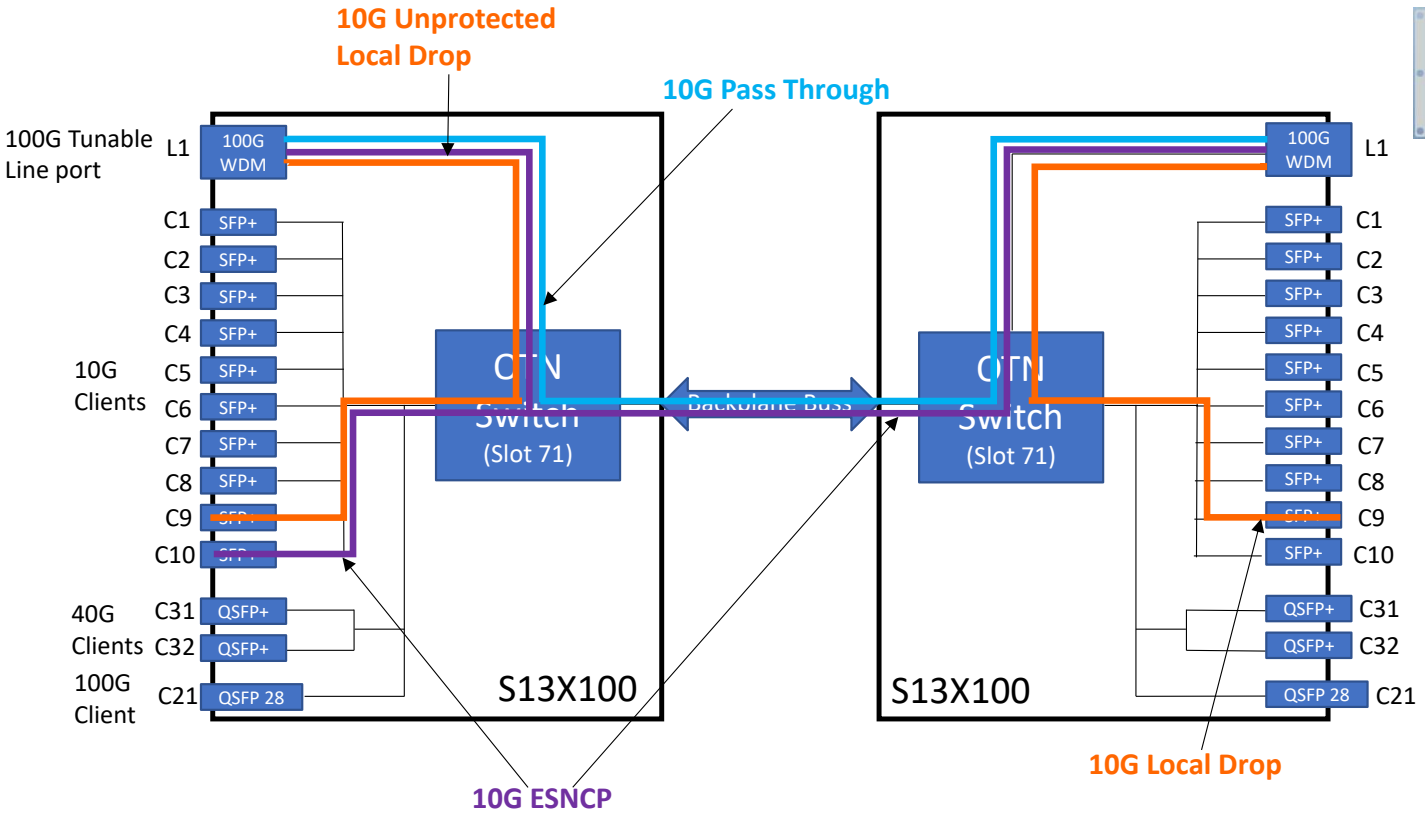
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S13X100 embedded OTN Switch with Backplane Buss

Enabling remote provisionable grooming and switching from Line-Line and Client to Line

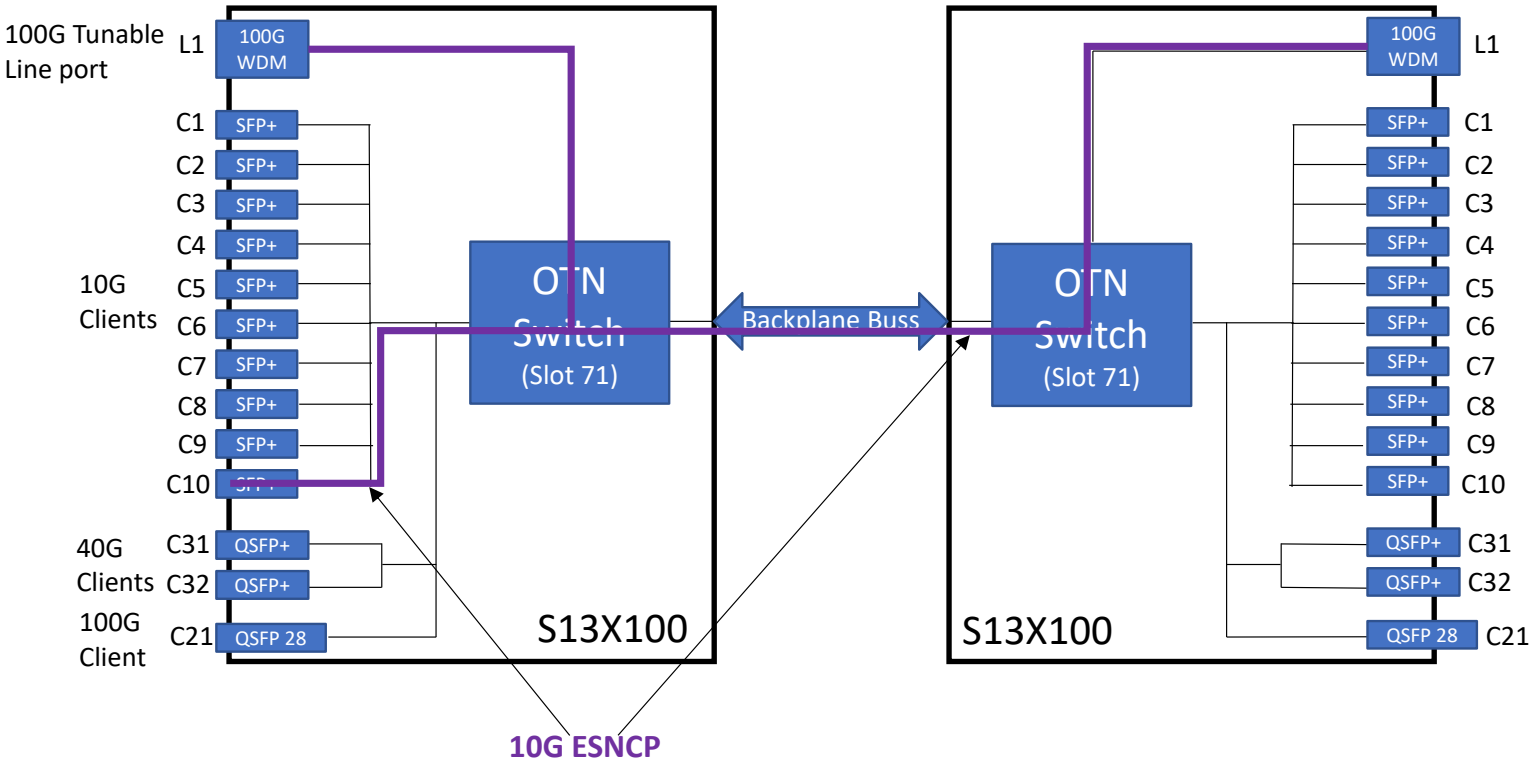
1830 PSS-8





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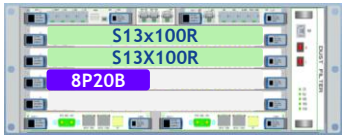
S13X100 10G ESNCP



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S13X100 ESNCP Sub-Rate

1830 PSS-8



100G Tunable Line port

10G Clients

40G Clients

100G Client

S13X100

OTN Switch (Slot 71)

Backplane Buss

10G =8xODU-0 (GbE)

ESNCP

Sub-rate Client i.e. GbE

8P20B

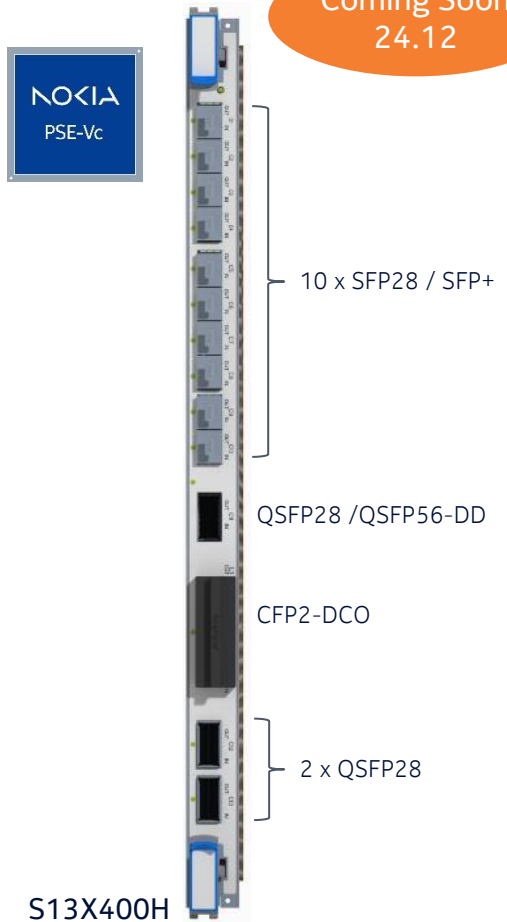
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S13X400H Muxponder

Unit Name	S13X400H
Part #	3KC72184AA
Size	Full Height, 1-slot, PSS8 / PSS16ii

	Features
Line	• 1 x CFP2-DCO WDM line port • 100G - 400G (QPSK / 8QAM / 16QAM modulation) • 28 – 63 Gbaud symbol rate • Nokia PSE-Vc DSP • Nokia SDFEC-G2 / CFEC+ / SCFEC
Client	• 10 x SFP28 / SFP+: 10 GE/25GE, OTU2/2e, FC16G/32G • 2 x QSFP28 100GE/OTU4 • 1 x QSFP28/QSFP56-DD 100GE/OTU4, 400GE
Features	• Integrated OTN switch • Standalone or mated (ie ADM) applications • OTN overhead (GCC, delay measurement, etc.) • GCC0 Line/Client • PTP / SyncE clock recovery / distribution
Protection	• OPSB5 • OPSUM • 1+1 OLP/OMSP • ESNCP when in ADM mode.
Power	• ~ 195 W (typical)

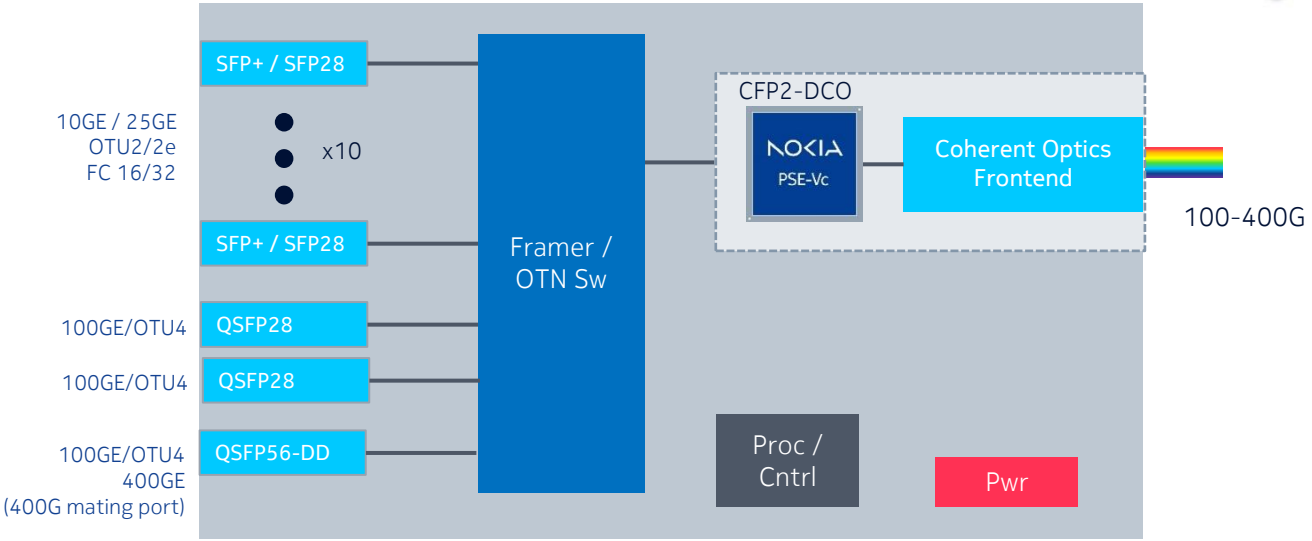
Coming Soon
24.12



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S13X400H Block Diagram

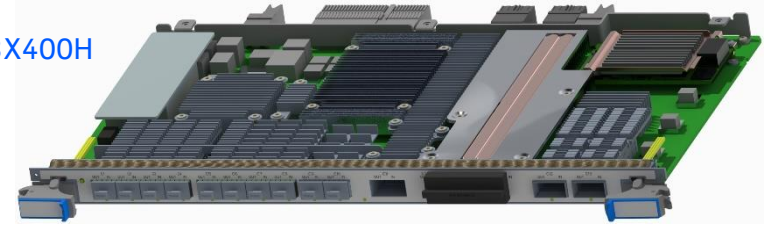




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S13X400H Applications – Flexible 400G Muxponder

S13X400H

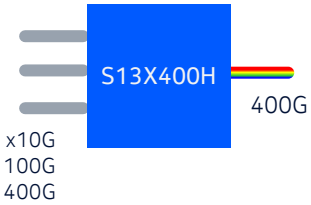


High Integration
Flexible multi-service

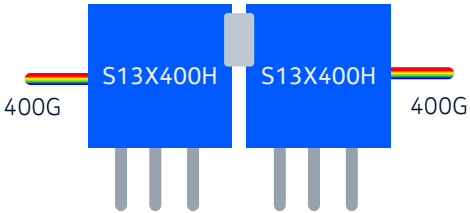
NOKIA
PSE-Vc

10G to 400G
client services

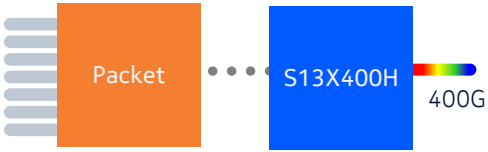
Muxponder



ADM



Uplink



(future)

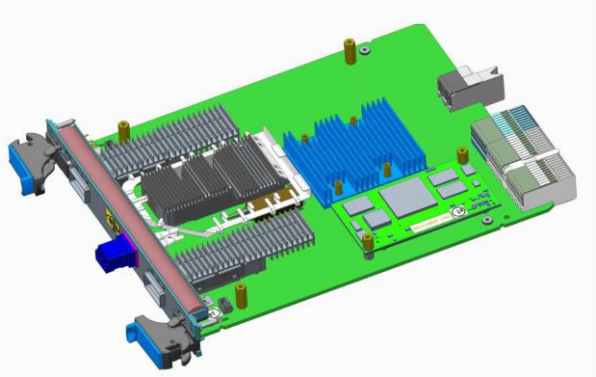
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S2AD200 – Compact 200G

Unit Name	S2AD200H
Part #	3KC69993AA
Size	1/2 Height, 1-slot

NOKIA PSE-3c

	Features
Line	• 100G QPSK or 200G 16QAM (provisioned) • Nokia PSE-3c coherent DSP • SD-FEC (25%) or SC-FEC (hard decision, staircase FEC, 7%)
Client	• 2 x QSFP28 100 GigE (LR4 / CWDM4 / SR4) OTU4
Features	• OTN overhead (GCC, delay measurement, etc.) • LLDP snooping
OAM&P	• OTN overhead (GCC, delay measurement, etc.) • O-SNCP client protection (OPSB) • OCH Protection
Power	• ~ 65 W (typical)





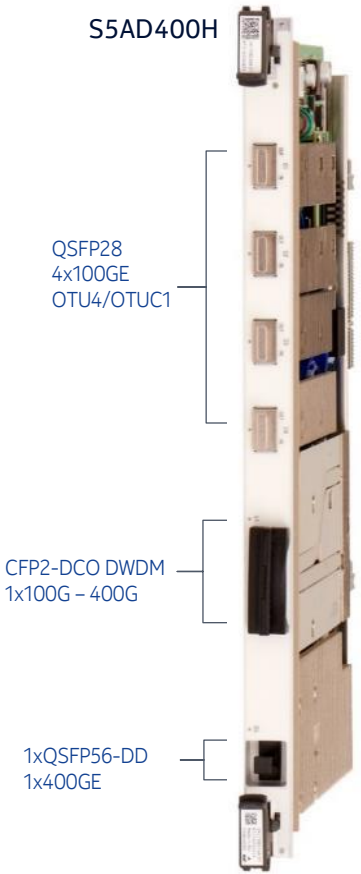
S2AD200

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NOKIA

S5AD400H – 400GE/4x100G Muxponder

Unit Name	S5AD400H
Part #	3KC70803AA
Size	Full Height, 1-slot

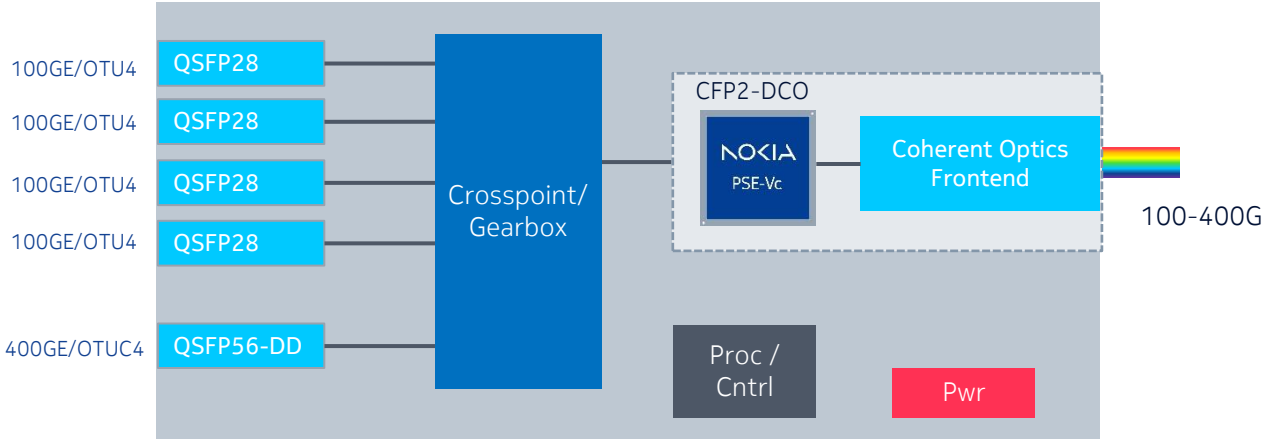
	Features
Line	• 1 x CFP2-DCO WDM line port • 100G - 400G (QPSK / 8QAM / 16QAM modulation) • 28 – 63 Gbaud symbol rate • Nokia PSE-Vc DSP • Nokia SDFEC-G2 / CFEC / CFEC+ / SCFEC supported (9.4 – 11.7dB NCG)
Client	• 4 x QSFP28 100GE/OTU4 • 1 x QSFP56-DD 400GE
Performance	• Optical Reach 400ZR+ over metro/regional, 100G/200G LH • CD 48,000 ps/nm • DGD 30 ps
Features	• 4x100GE Muxponder or 400GE (client) Transponder • GCC0 remote management • LLDP snooping • L0 Restoration (GMPLS)
Protection Options	• OSNCP (OPSB) Client • OCH Protection
Power	• ~ 100 W (typical)



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S5AD400H Block Diagram





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nx100GE or 400GE High Capacity Transport Application



High Capacity Data Center Interconnect

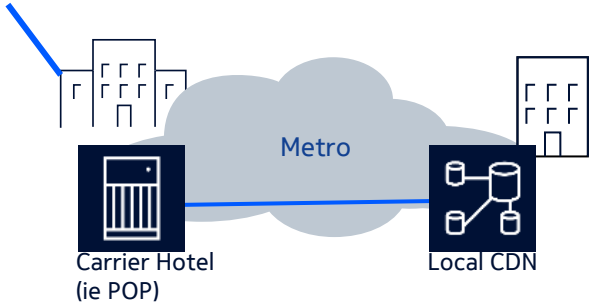
- Data center interconnect using nx100GE or 400GE
- Supported on 1830 PSS-32/16ii/8 shelves
- Supported configurations
  - Pt – Pt Mux/Demux (FOADM)
  - ROADM (classic, CDC-F, C-F)

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DCI Metro – Application

Metro DCI

- ICPs building small, local caching sites
- Metro DCI connects local CDN to metro POP (ie carrier hotel)
- Typical attributes
  - < 50 km
  - n x 100GE / 400GE
- Ideal application for S5AD400H





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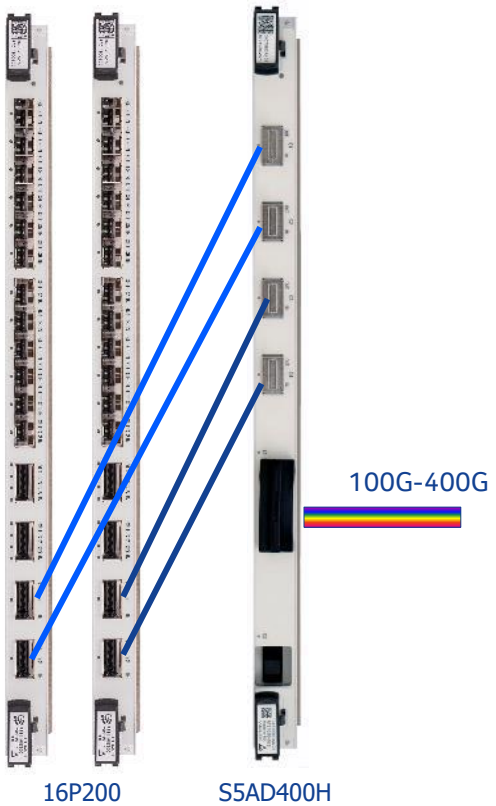
High Capacity nx10GAggregation

Low Speed Aggregation

- Pair S5AD400H with one or two 16P200 to support low speed aggregation applications
- Units interconnected via front panel QSFP28 OTU4 links

16P200 – S5AD400H Aggregation

- Support up to (12) x 10GE per 16P200
- Multi-Service SFP28 ports
  - 10GE
  - OTU2/2e
  - OC-192/STM-64
  - 8/10/16/32 FC

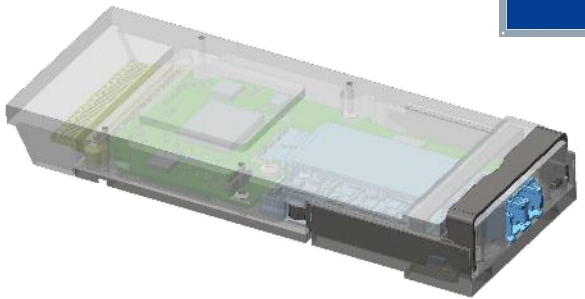


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C2DCO4 CFP2-DCO Coherent Module

Unit Name	C2DCO4 / C2DCO4 Hi Perf (R24.6)
Part #	3KC93809AA / 3KC93911AA
Size	CFP2

	Features
Line	• 1 x CFP2-DCO WDM line port • 100G - 400G (QPSK / 8QAM / 16QAM modulation) • 28 – 63 Gbaud symbol rate • Nokia SDFEC-G2 (C2DCO4) • C2DCO4 High Performance version (3KC93911AA) • OpenROADM (C2DCO4O) oFEC option (3KC93826AA)
Features	• 400ZR+ target application • Interoperability (ie compatible existing Nokia coherent transponders) • Integrated OTN mapper/framer • Non-encryption
Power	• < 27 W (typical)



C2DCO4



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C2DCO4 High Performance Version

NEW
24.6

C2DCO4 High Performance version (3KC93911AA)

- Higher output power, up to +3 dBm

	QPSK	8QAM	16QAM
Std	-1 dBm	-2 dBm	-2 dBm
High Perf	+3 dBm	+2 dBm	+2 dBm



C2DCO4 High Performance

- Improved OSNR performance
  - ~0.5 dB OSNR improvement
    - OSNR improvement ranges from 0.3 to 0.8 dB, dependent on carrier profile

- Supported cards (R24.6)
  - S5AD400H 1830 PSS
  - DD2M4 1830 PSI-M

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S6AD600H 600G Muxponder Module

Unit Name	S6AD600H / S6AD600L / S6AD600E (R24.2H)
Part #	3KC71192AA / 3KC71722AA / 3KC71270AA
Size	1/2-height, 2-slot

	Features
Line	• 100G – 600G WDM line port (QPSK – QAM16, PCS) • 33 – 90 Gbaud symbol rate • Nokia PSE-Vs DSP • Probabilistic Constellation Shaping (PCS) • Nokia SDFEC (15%, 25%)
Client	• 5 x QSFP28 100GE/OTU4 • 1 x QSFP28/QSFP56-DD 100GE/400GE
Applications	• High performance metro – ULH distances • 6x100GE/OTU4 Muxponder • 400GE Transponder • Mixed 100GE / 400GE transport
Power	• ~ 130W (typical)



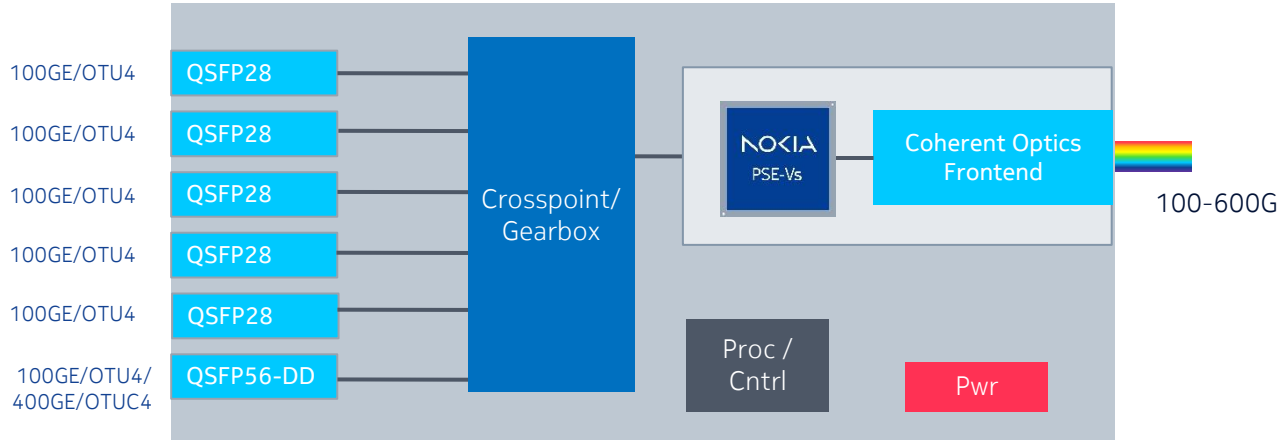
S6AD600H

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S6AD600H Block Diagram



S6AD600H



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D6AD2T4H 1830 PSS-16ii Muxponder Card

NEW 24.6

NOKIA PSE-6s

Unit Name	D6AD2T4H / D6AD2T4L (R24.2H)
Part #	3KC71964AA / 3KC72384AA
Size	Double-width (-16ii shelf)

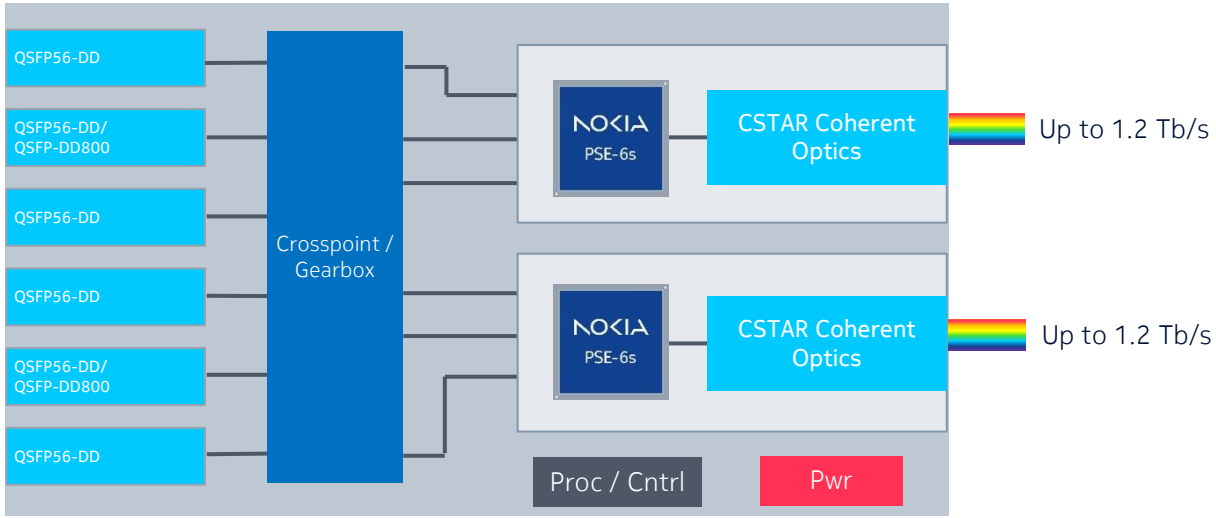
	Features
Line	• (2) x 1.2 Tb/s WDM line ports, 130Gbaud • 3rd Gen Probabilistic Constellation Shaping (PCS), Shaped-QAM64, Shaped-QAM16 • Nokia SDFEC (15%, 25%, variable) • Continuous baud rate flexible profiles • 0.1 GHz central frequency tuning
Client	• 4 x QSFP56-DD 400GE (100GE/OTU4 via breakout cable) • 2 x QSFP56-DD/QSFP800 400GE/800GE (100GE/OTU4 via breakout cable)
Applications	• 1.2Tb/s Metro / 800Gbps long haul / 600G ultra-long haul • 2.4 Tb/s muxponder (n x 100G/400G/800G) • Mixed 100G / 400G / 800G client traffic transport
Protection	• OSNCP (OPSB5) • OCHP (OPSUM) • OLP/OMSP (OPSA)
Features	• Integrated test signal • Comprehensive optical & digital PMs • Client/line GCC0 processing • LLDP snooping



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NOKIA

D6AD2T4H Block Diagram

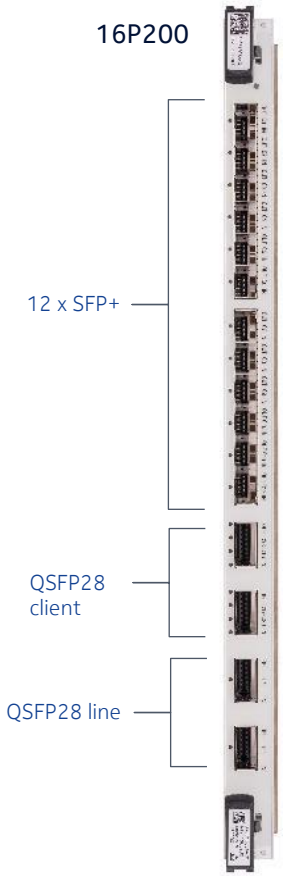


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16P200 – nx10G Muxponder

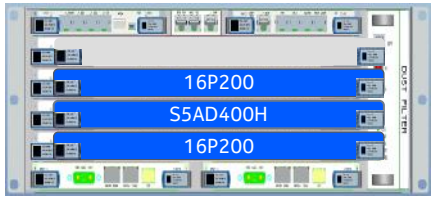
Unit Name	16P200
Part #	3KC71203AA
Size	Full Height, 1-slot

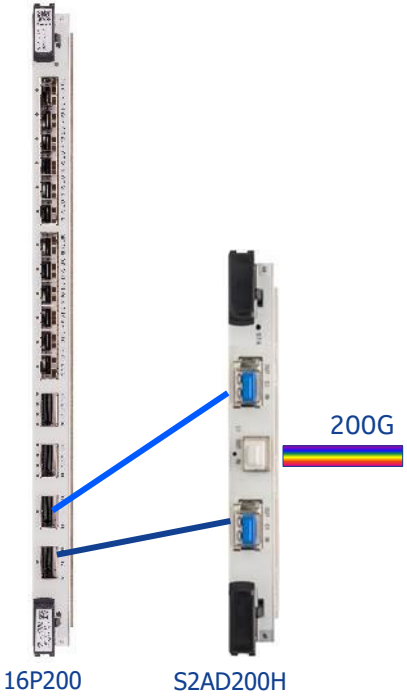
	Features
I/O Ports	• 12 x SFP+/SFP28 • 10GE, OTU2/2e, OC-192/STM64, FC-8G /10G/16G/32G • 2 x QSFP+/QSFP28 clients • 100GE, 4x10GE • 2 x QSFP28 line • OTU4
Applications	• nx10G Muxponder • Paired with transponder cards, 100G-600G WDM line interface • S2AD200H • S5AD400H • S6AD600
Features	• Multi-protocol nx10G client ports • GCC0 remote management • Client/Line loopbacks with PRBS test signal injection / monitoring • NetConf/Yang
Power	• ~ 90 W (typical)

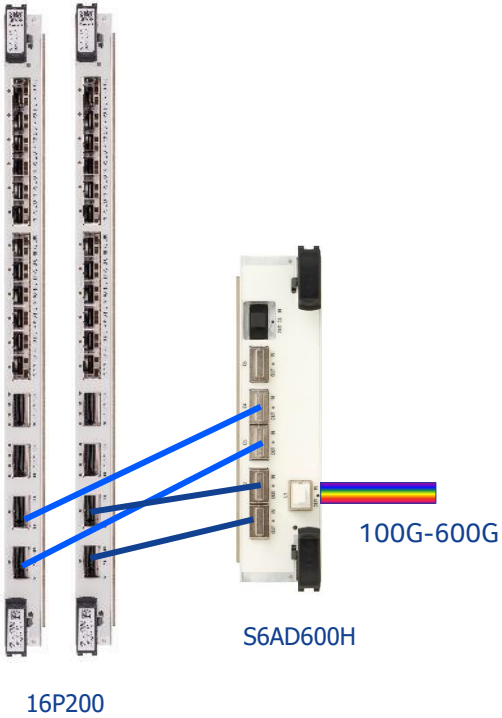


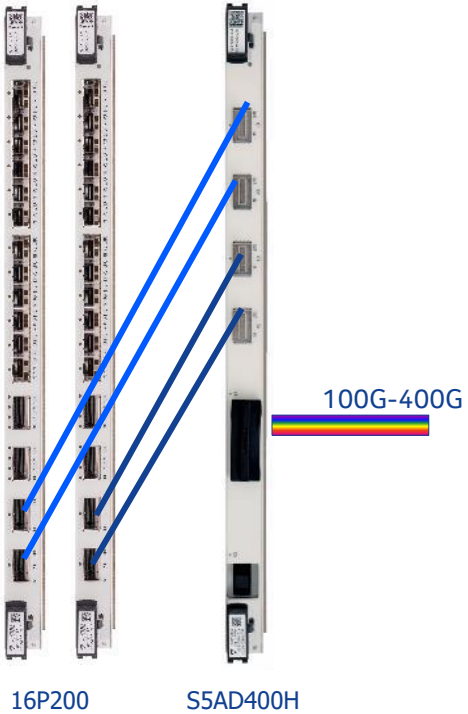
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Application Examples










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